WAITLESS
DIGITAL PATIENT FLOW MANAGEMENT

SYSTEM DEVELOPMENT AMENDMENTS

WORLD-CLASS FEATURE REQUIREMENTS

Detailed design specification for a modern, secure and scalable hospital queue management platform.

OPD-025
REGISTRATION

LAB-014
DIAGNOSTICS

PHARM-041
PHARMACY

21	82	4	3
CORE MODULES	DETAILED FEATURES	TRIAGE LEVELS	DEPLOYMENT MODELS

FEATURE REQUIREMENTS & SYSTEM DESIGN DOCUMENT
Version 1.0 | July 2026

DOCUMENT OVERVIEW

A complete blueprint for the WaitLess platform

This designed specification organises the proposed capabilities into a clear, implementation-ready structure. It covers the complete patient journey - from registration and triage to departmental routing, notifications, reporting, privacy, offline operation and advanced intelligence.

21	82	4	3
CORE MODULES	DETAILED FEATURES	TRIAGE PRIORITIES	DEPLOYMENT OPTIONS

DESIGN PILLARS

01 Patient-centred access Digital registration, self-service and clear patient guidance.	**02 Clinical prioritisation** Colour-coded triage, reassessment and emergency fast-tracking.	**03 Real-time operations** Live queues, routing, alerts and department dashboards.
04 Secure governance Role-based access, confidentiality and complete audit trails.	**05 Resilient deployment** Local, offline, low-bandwidth and hybrid operation.	**06 Intelligent improvement** Analytics, prediction, bottleneck detection and AI support.

SCOPE & INTENT

WaitLess is envisioned as a configurable digital patient-flow platform for public and private health facilities. The design prioritises shorter waiting times, fair clinical prioritisation, transparent queues, better patient communication, operational visibility and dependable use in resource-constrained environments.

> **DESIGN PRINCIPLE** **Technology should support clinical judgement and hospital policy - not replace qualified healthcare decision-making.**

DOCUMENT MAP

Contents at a glance

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Reading guide: Each numbered feature contains a functional description followed by the operational value it delivers. Tables summarise rules, routing and access responsibilities.

01

PART I

Patient Access & Clinical Flow

Modules 1-4: registration, self-service, triage and queue management

MODULE 01

Patient Registration

1.1 Electronic Patient Registration

This feature allows hospital reception staff to capture patient details digitally instead of writing them manually in books or paper forms. The receptionist enters information such as patient name, age, gender, national ID number, contact number, address, next of kin, and reason for visit. Once the patient is registered, the system automatically creates a visit record and assigns the patient to the correct hospital workflow.

WHY IT MATTERS This feature reduces paperwork, prevents loss of records, improves patient identification, and allows the hospital to track every patient from entry to exit.

1.2 Returning Patient Search

This feature allows staff to search for patients who have visited the hospital before. A patient can be searched using their name, phone number, national ID, patient number, or QR code. When the patient is found, the system displays their previous visit history and allows staff to create a new visit without re-entering all details.

WHY IT MATTERS This saves time during registration and reduces duplicate patient records.

1.3 Patient Token Number Generation

After registration, the system automatically generates a unique queue/token number for the patient. This number is used to identify the patient in the queue without publicly showing their full name. For example, a patient may receive a number such as **OPD-025**, **LAB-014**, or **PHARM-041**.

The token number can be printed, displayed on the patient's phone, sent through WhatsApp, or shown on queue display screens.

1.4 QR Code Patient Card

The system can generate a QR code for each patient or visit. Staff can scan this QR code to quickly retrieve the patient's record, current queue position, department, and visit status. The QR code may be printed on a small ticket or stored on the patient's phone.

WHY IT MATTERS This makes check-in faster and reduces typing errors.

1.5 Patient Category Identification

During registration, staff can classify patients into categories such as walk-in patient, referred patient, emergency patient, elderly patient, pregnant patient, child patient, disabled patient, or chronic care patient. This helps the system apply the correct queue rules and priority levels.

For example, elderly patients, pregnant women, or emergency cases may be prioritised according to hospital policy.

MODULE 02

Patient Self-Service

2.1 Online Pre-Registration

This feature allows patients to register before arriving at the hospital using a web portal. The patient enters their basic details, selects the service they need, and receives a pre-registration confirmation. When they arrive at the hospital, staff only need to verify the details and activate the visit.

WHY IT MATTERS This reduces congestion at the registration desk and shortens the time spent at the hospital.

2.2 Self Check-In

Patients who have registered online can check themselves in when they arrive at the hospital. They can do this by scanning a QR code, entering their phone number, or using a kiosk. Once checked in, the system places them in the correct queue.

WHY IT MATTERS This feature is useful in busy hospitals because it reduces pressure on reception staff.

2.3 Patient Queue Status Page

The patient can view their current queue status from a phone or web browser. The page shows their token number, current department, queue position, estimated waiting time, and whether their turn is approaching.

WHY IT MATTERS This gives patients more control and reduces the need to constantly ask staff, "When am I being served?"

2.4 Patient Feedback Form

After service, the system can ask the patient to rate their hospital experience. The patient may rate waiting time, staff behaviour, communication, cleanliness, and overall satisfaction. The feedback is stored and used by hospital management to improve service quality.

WHY IT MATTERS This helps the hospital understand patient complaints and satisfaction trends.

MODULE 03

Triage

The proposal identifies triage as a key requirement because Chinhoyi Provincial Hospital uses a colour-coded triage system: Red, Yellow, Green, and Black.

3.1 Digital Triage Assessment

This feature allows nurses or triage officers to assess patients digitally. The staff member records the patient's symptoms, vital signs, pain level, condition severity, and emergency indicators. The system then stores the triage assessment and assigns the patient to the correct priority category.

WHY IT MATTERS This replaces handwritten triage records and gives the hospital a digital history of patient assessments.

3.2 Colour-Coded Triage Priority

The system uses colour-coded priority levels to classify patients according to severity:

Colour	Meaning	How It Works
Red	Critical emergency	Patient is sent immediately to emergency/resuscitation area
Yellow	Serious but not immediately life-threatening	Patient is prioritised above normal OPD cases
Green	Minor or stable case	Patient joins the normal waiting queue
Black	Deceased case	Patient is handled according to hospital mortuary procedure

WHY IT MATTERS This ensures that critically ill patients are not treated the same as minor cases.

3.3 Automatic Emergency Fast-Tracking

When a patient is marked as Red or high-risk, the system automatically pushes them to the top of the emergency queue. Staff on the casualty or emergency dashboard receive an alert that a critical patient is waiting.

WHY IT MATTERS This improves emergency response and reduces the risk of critically ill patients being delayed.

3.4 Triage Reassessment Alerts

If a patient waits too long after being triaged, the system can alert nurses to reassess the patient. For example, a Yellow patient who has waited for 30 minutes may need to be checked again to ensure their condition has not worsened.

WHY IT MATTERS This feature improves patient safety because some patients deteriorate while waiting.

3.5 Manual Triage Override

Although the system can recommend a triage level, trained medical staff must be able to override it. For example, the system may suggest Green, but the nurse may decide the patient should be Yellow based on clinical judgement.

WHY IT MATTERS This keeps medical decision-making under qualified staff control.

MODULE 04

Queue Management

4.1 Automatic Queue Assignment

After registration and triage, the system automatically places the patient in the correct queue. For example, the patient may be sent to OPD, Casualty, Laboratory, Radiology, Pharmacy, or OI Clinic depending on their reason for visit and triage level.

WHY IT MATTERS This removes confusion and helps patients follow the correct hospital pathway.

4.2 Department-Specific Queues

Each department has its own queue. OPD, Pharmacy, Laboratory, Radiology, Casualty, and OI Clinic can each manage their own list of waiting patients. Staff in each department only see the patients relevant to their work.

WHY IT MATTERS This makes the system organised and prevents one general queue from becoming confusing.

4.3 Call Next Patient

Staff can click a **Call Next Patient** button to call the next person in the queue. The system selects the correct patient based on priority, arrival time, and department rules. Once called, the patient's status changes from "Waiting" to "Called" or "In Service."

WHY IT MATTERS This makes patient calling faster, fairer, and easier to track.

4.4 Missed Turn Handling

If a patient is called but does not appear, staff can mark them as **Missed Turn** or **Absent**. The system can then send a notification to the patient and temporarily move them aside without deleting them from the queue.

WHY IT MATTERS This solves the problem of patients leaving the waiting area and missing their turn.

4.5 Patient Recall

If a patient misses their turn, staff can recall them later. The system can place the patient back into the queue according to hospital rules. For example, they may be placed after the next three patients instead of going to the very back.

WHY IT MATTERS This keeps the queue fair while still allowing flexibility.

4.6 Queue Transfer

Patients often move between departments. For example, after seeing a doctor, a patient may be sent to Laboratory, then back to the doctor, then to Pharmacy. The queue transfer feature allows staff to move the patient digitally from one department to another.

WHY IT MATTERS This helps track the full patient journey from registration to discharge.

4.7 Waiting Time Estimation

The system calculates estimated waiting time based on the number of patients in the queue, average service time, number of available staff, and patient priority level. The estimated time can be shown to staff and patients.

WHY IT MATTERS This improves transparency and helps patients plan their time.

4.8 Bottleneck Detection

A bottleneck happens when patients are stuck for too long in one department. The system can detect this by comparing waiting times across departments. For example, if Laboratory has many patients waiting longer than expected, the dashboard can show a warning.

WHY IT MATTERS This helps hospital managers identify departments that need more staff or resources.

02

PART II

Department Operations & Communication

Modules 5-8: routing, dashboards, notifications and display screens

MODULE 05

Department Routing

The proposal explains that the current hospital flow includes registration, nurse observation, consultation, laboratory, radiology, pharmacy, treatment room, and departure, but the process is paper-based and difficult to monitor digitally.

5.1 Automatic Department Routing

After a patient is assessed, the system directs them to the next appropriate department. For example:

Patient Need	System Routing
Minor illness	OPD queue
Emergency case	Casualty queue
Lab test needed	Laboratory queue
X-ray needed	Radiology queue
Medicine collection	Pharmacy queue
HIV/TB/ART services	OI Clinic queue

WHY IT MATTERS This prevents patients from being sent to the wrong place.

5.2 Doctor-to-Lab Routing

When a doctor requests a lab test, the system transfers the patient to the Laboratory queue. The lab staff can see the request and mark the test as pending, in progress, or completed. Once results are ready, the patient can be routed back to the doctor.

WHY IT MATTERS This avoids physical movement of papers between departments.

5.3 Doctor-to-Pharmacy Routing

After consultation, the doctor can send the patient to Pharmacy. The pharmacy receives the patient in its queue and prepares medication. When medicine is ready, the system can notify the patient.

WHY IT MATTERS This makes the medicine collection process smoother.

5.4 OI Clinic Routing

The OI Clinic handles multiple service streams such as ART resupply, TB services, viral load testing, counselling, and cervical cancer screening. The routing feature separates these service types so that patients are sent to the correct queue.

WHY IT MATTERS This reduces confusion and missed appointments in chronic care services.

MODULE 06

Real-Time Staff Dashboard

6.1 Live Queue Dashboard

The live dashboard shows current queue activity across departments. Staff can see how many patients are waiting, who is being served, who has missed their turn, and which department is congested.

WHY IT MATTERS This gives staff real-time visibility instead of relying on paper lists.

6.2 Department Dashboard

Each department has its own dashboard. For example, Pharmacy staff only see patients waiting for medicine, while Laboratory staff only see patients waiting for tests. The dashboard can show queue number, patient priority, waiting time, and service status.

WHY IT MATTERS This keeps each department focused on its own workflow.

6.3 Admin Dashboard

The admin dashboard gives hospital managers an overview of the whole hospital. It shows total patients registered, average waiting time, number of emergency cases, number of completed visits, no-shows, and department performance.

WHY IT MATTERS This helps management make evidence-based decisions.

6.4 Real-Time Alerts

The dashboard can show alerts when something needs attention. Examples include:

Alert	Meaning
Critical patient waiting	Red triage patient needs immediate attention
Queue too long	A department has too many waiting patients
Patient waited too long	A patient has exceeded safe waiting time
Notification failed	WhatsApp or web push message failed
Staff unavailable	A department has no active staff serving

WHY IT MATTERS This makes the system proactive instead of passive.

MODULE 07

Patient Notification

Your proposal identifies notifications as one of the most important features. It states that patients can be notified through WhatsApp and web push notifications when their turn is approaching, especially if they temporarily leave the hospital premises.

7.1 WhatsApp Queue Notifications

When a patient registers with a WhatsApp number, the system can send queue updates through WhatsApp. The patient may receive messages such as:

- “Your token number is OPD-025.”
- “You are number 3 in the queue.”
- “Please return to OPD. Your turn is approaching.”
- “Your lab results are ready. Please return to the doctor.”

WHY IT MATTERS This is useful because many patients already use WhatsApp and do not need to install a new app.

7.2 Web Push Notifications

Patients who register through the web application can receive browser notifications. These notifications appear on their phone or computer even if they are not actively looking at the page.

> **WHY IT MATTERS** This provides a low-cost alternative to SMS and supports patients who use the web portal.

7.3 Notification Retry System

Sometimes notifications may fail because of poor internet connection or network problems. The system should store failed notifications and retry sending them when the connection improves.

> **WHY IT MATTERS** This is important because the proposal mentions that the system should be suitable for low-bandwidth and resource-limited environments.

7.4 Notification Log

Every notification sent by the system should be recorded. The log should show the patient, message type, time sent, delivery status, and whether it succeeded or failed.

> **WHY IT MATTERS** This helps staff prove that a patient was notified and helps administrators troubleshoot communication problems.

7.5 Missed Turn Notification

If a patient is called and does not appear, the system sends a message telling them that they missed their turn and should return to the department. This reduces queue disruption and allows staff to manage absent patients fairly.

MODULE 08

Queue Display Screens

8.1 Waiting Area Display Board

This feature displays live queue numbers on a TV or monitor in the waiting area. Patients can see which number is currently being served and which numbers are coming next.

Although the proposal lists queue display monitors as a future enhancement, it is a very strong world-class feature because it supports patients who do not have smartphones or internet access.

8.2 Department-Specific Display Boards

Each department can have its own display screen. For example, OPD displays OPD tokens, Pharmacy displays Pharmacy tokens, and Laboratory displays Lab tokens.

> **WHY IT MATTERS** This avoids confusion because patients only follow the queue board relevant to their department.

8.3 Audio Calling

The system can call out patient token numbers using speakers. For example, it may announce: “OPD zero two five, please proceed to consultation room two.”

> **WHY IT MATTERS** This is useful for patients who may not be watching the display screen.

03

PART III

Reporting, Analytics & Governance

Modules 9-12: reporting, decision support, security and confidentiality

MODULE 09

Reporting

9.1 Daily Patient Report

This report shows the number of patients served in a day, broken down by department, service type, gender, age group, and triage category. Hospital managers can use this report to understand daily workload.

9.2 Average Waiting Time Report

This report calculates how long patients waited in each department. For example, it may show that OPD patients waited an average of 2 hours, while Pharmacy patients waited 45 minutes.

WHY IT MATTERS This helps management identify where delays are happening.

9.3 Department Performance Report

This report compares departments based on patients served, average service time, waiting time, and queue length. It helps hospital administrators know which departments are performing well and which need improvement.

9.4 No-Show Report

This report shows how many patients missed their turn or left before being served. It can also show whether notifications helped reduce no-shows.

WHY IT MATTERS This is important because the proposal identifies no-shows as a major problem when patients leave the hospital premises while waiting.

9.5 Triage Report

This report shows how many patients were classified as Red, Yellow, Green, or Black. It helps the hospital understand emergency demand and plan staffing for casualty and OPD.

9.6 Export to PDF, Excel, and CSV

Hospital managers may need reports for meetings, audits, or Ministry of Health submissions. The system should allow reports to be exported as PDF, Excel, or CSV files.

WHY IT MATTERS This makes the system more professional and useful for administration.

MODULE 10

Analytics and Decision Support

10.1 Peak Hour Analysis

The system can analyse patient arrival times and identify the busiest hours of the day. For example, it may show that most patients arrive between 7:00 AM and 10:00 AM.

WHY IT MATTERS This helps managers schedule more staff during busy periods.

10.2 Queue Congestion Prediction

Using previous queue data, the system can predict when a department is likely to become congested. For example, if Pharmacy usually becomes busy after OPD consultations, the system can warn staff early.

WHY IT MATTERS This improves planning and reduces long waits.

10.3 Resource Allocation Suggestions

The system can suggest where additional staff may be needed. For example, if Laboratory has many patients waiting and OPD is less busy, management may temporarily assign support staff to Laboratory.

WHY IT MATTERS This helps improve hospital efficiency.

MODULE 11

Security and Access Control

11.1 Role-Based Access Control

Different users should access different parts of the system depending on their roles. For example:

User Role	Access
Receptionist	Patient registration
Nurse	Triage and patient assessment
Doctor	Consultation queue and referrals
Pharmacist	Pharmacy queue
Lab staff	Laboratory queue
Administrator	Users, reports, settings
Hospital manager	Analytics and performance reports

WHY IT MATTERS This protects patient information and prevents unauthorised actions.

11.2 Secure Login

Every staff member logs in using their own username and password. The system records who performed each action, such as registering a patient, changing triage level, or marking a patient as served.

WHY IT MATTERS This creates accountability.

11.3 Audit Trail

The audit trail records important system activities. For example, it records who changed a patient's priority, who transferred a patient, who deleted a record, or who exported a report.

WHY IT MATTERS This improves transparency and helps investigate mistakes or misuse.

11.4 Automatic Logout

If a staff member leaves the computer idle, the system automatically logs them out after a set period. This prevents other people from accessing patient records on unattended computers.

MODULE 12

Privacy and Patient Confidentiality

12.1 Anonymous Queue Display

Instead of showing patient names on public screens, the system displays token numbers. For example, it shows **OPD-025** instead of **Norris Marava**.

WHY IT MATTERS This protects patient privacy.

12.2 Consent for Notifications

Before sending WhatsApp or web push notifications, the system should ask the patient for consent. The patient must agree that the hospital can send queue updates to their phone.

WHY IT MATTERS This supports ethical handling of patient communication.

12.3 Confidential Clinic Mode

For sensitive departments such as OI Clinic, the system should avoid exposing patient names, diagnosis, or service type publicly. It should use private token numbers and limit access to authorised staff only.

WHY IT MATTERS This protects patients who may be receiving HIV, TB, counselling, or other sensitive services.

04

PART IV

Resilience, Scheduling & Clinical Services

Modules 13-16: offline readiness, appointments, OI Clinic and Pharmacy

MODULE 13

Offline and Low-Bandwidth Operation

The proposal states that the system should be designed for low-bandwidth networks and local hospital infrastructure.

13.1 Local Area Network Hosting

The system can be hosted on a local hospital server so that staff can continue using it even when internet connection is poor. Computers inside the hospital connect to the system through the local network.

WHY IT MATTERS This is important for public hospitals where internet may be unstable.

13.2 Offline Queue Management

If the internet goes down, the core queue functions should still work locally. Staff should still be able to register patients, call patients, transfer patients, and complete visits.

WHY IT MATTERS Only internet-dependent features such as WhatsApp notifications may wait until internet returns.

13.3 Queued Notifications

When internet is unavailable, WhatsApp and web push messages are stored in a pending queue. Once connectivity returns, the system automatically sends the pending messages.

WHY IT MATTERS This ensures communication is not completely lost during network problems.

13.4 Auto-Save Forms

If a staff member is entering patient details and the connection becomes unstable, the system should auto-save the form locally. This prevents data loss and avoids forcing staff to re-enter the same information.

MODULE 14

Appointment Scheduling

14.1 Online Appointment Booking

Patients can book appointments online for services such as OPD review, OI Clinic, ART resupply, TB review, or follow-up consultation. The system assigns a date, time slot, and queue category.

WHY IT MATTERS This helps reduce overcrowding caused by too many walk-in patients arriving at the same time.

14.2 Follow-Up Appointment Scheduling

After consultation, the doctor or nurse can schedule the patient's next visit. The patient receives a notification confirming the appointment date and time.

WHY IT MATTERS This is useful for chronic patients who need regular care.

14.3 Appointment Reminders

The system sends reminders before the appointment date. This may be done through WhatsApp, web push, SMS, or email.

WHY IT MATTERS This reduces missed appointments.

MODULE 15

OI Clinic Features

15.1 Multiple Service Stream Management

The OI Clinic may handle ART resupply, TB services, viral load testing, counselling, and cervical cancer screening. The system separates these services into different queues so that each patient follows the correct path.

WHY IT MATTERS This prevents confusion and makes chronic care management more organised.

15.2 Confidential Patient Handling

Because OI Clinic services can be sensitive, the system should protect patient privacy by using token numbers, limited staff access, and confidential service labels.

WHY IT MATTERS This builds patient trust and protects dignity.

15.3 Missed Appointment Tracking

If a chronic care patient misses an appointment, the system records it and can notify staff for follow-up. The system may also send the patient a reminder to reschedule.

WHY IT MATTERS This improves continuity of care.

MODULE 16

Pharmacy Features

16.1 Pharmacy Queue

Patients sent to Pharmacy are placed in a pharmacy-specific queue. Pharmacy staff can see waiting patients, prepare medication, and mark prescriptions as ready or collected.

WHY IT MATTERS This improves the medicine collection process.

16.2 Prescription Ready Notification

When medicine is ready, the patient receives a message telling them to proceed to Pharmacy. This means the patient does not need to stand near the pharmacy counter the whole time.

WHY IT MATTERS This reduces crowding around Pharmacy.

16.3 Medicine Out-of-Stock Flag

If a prescribed medicine is not available, pharmacy staff can mark it as out of stock. The system records this and can notify the doctor or patient.

WHY IT MATTERS This improves communication and prevents patients from waiting unnecessarily.

05

PART V

Diagnostics, Workforce & Patient Experience

Modules 17-20: diagnostics, staff management, patient experience and global scale

MODULE 17

Laboratory and Radiology Features

17.1 Lab Test Queue

When a doctor requests a lab test, the patient is placed in the Laboratory queue. Lab staff can update the test status as pending, sample collected, in progress, or completed.

WHY IT MATTERS This helps doctors and patients know the progress of lab requests.

17.2 Results Ready Alert

When lab or radiology results are ready, the system can notify the doctor and route the patient back to the doctor's review queue.

WHY IT MATTERS This reduces delays caused by patients waiting without knowing that results are ready.

17.3 Turnaround Time Tracking

The system records how long it takes from test request to result completion. This helps management identify delays in Laboratory or Radiology.

MODULE 18

Staff Management Features

18.1 Staff Profiles

Each staff member has a profile with name, role, department, contact details, and login credentials. This helps administrators manage system users.

18.2 Department Assignment

Staff can be assigned to specific departments. For example, a nurse may be assigned to Triage, while a pharmacist is assigned to Pharmacy.

WHY IT MATTERS This ensures staff only see the queues and actions relevant to their duties.

18.3 Staff Workload Tracking

The system can show how many patients each staff member served during a shift. This helps managers understand workload distribution.

WHY IT MATTERS This should be used carefully and fairly, not to punish staff, but to improve planning.

MODULE 19

Patient Experience Features

19.1 Estimated Waiting Time Display

Patients can see how long they are likely to wait. This improves transparency and reduces anxiety because patients know what to expect.

19.2 Leave-and-Return Support

Patients may leave the waiting area temporarily and return when notified. This is one of the most important improvements because the proposal identifies no-shows and queue disruption as problems when patients leave without being notified.

19.3 Service Direction Guidance

The system can show instructions such as “Proceed to Laboratory,” “Return to Doctor Room 2,” or “Go to Pharmacy Counter 1.” This helps patients navigate the hospital more easily.

MODULE 20

Global / World-Class Features

20.1 Multi-Hospital Support

The system can be designed to support more than one hospital. Each hospital has its own departments, staff, queues, reports, and settings.

WHY IT MATTERS This allows WaitLess to grow from one hospital prototype into a national or international solution.

20.2 Multi-Language Support

The system can support English, Shona, Ndebele, and other languages. Staff and patients can choose their preferred language.

WHY IT MATTERS This makes the system more inclusive and easier to use.

20.3 Configurable Hospital Workflows

Different hospitals may have different patient flow processes. The system should allow administrators to configure departments, service types, queue rules, working hours, and triage policies.

WHY IT MATTERS This makes WaitLess adaptable globally.

20.4 Cloud, Local, or Hybrid Deployment

The system can support three deployment models:

Deployment Type	Description
Local	Runs on a hospital server/LAN
Cloud	Runs online and is accessed through the internet
Hybrid	Core system runs locally, with cloud backup and remote reporting

WHY IT MATTERS This makes the system suitable for both low-resource hospitals and advanced hospitals.

06

PART VI

Intelligent & Advanced Capabilities

Module 21: predictive analytics, AI bottleneck detection and patient chatbot

MODULE 21

AI and Advanced Features

21.1 AI Waiting Time Prediction

The system can use past queue data to predict waiting times more accurately. It considers number of patients, staff availability, department speed, and patient priority.

WHY IT MATTERS This makes waiting time estimates smarter.

21.2 AI Bottleneck Detection

The system can automatically identify when one department is slowing down the whole hospital flow. For example, if many patients are waiting for lab results, the system alerts management.

WHY IT MATTERS This helps hospitals respond before queues become too long.

21.3 AI Chatbot for Patients

A chatbot can answer common patient questions such as:

- “What is my queue number?”
- “How many people are before me?”
- “Where should I go next?”
- “What time is my appointment?”

WHY IT MATTERS This reduces pressure on hospital reception staff.

FINAL OVERVIEW

Executive Feature Summary

A concise description of the complete WaitLess platform

EXECUTIVE SUMMARY

WaitLess in one clear description

WaitLess will provide a complete digital patient flow management platform that supports electronic registration, triage prioritisation, department routing, real-time queue monitoring, patient notifications, staff dashboards, and management reporting. Patients will be registered digitally and assigned a queue number based on their service needs and triage priority. The system will then route them to the correct department, monitor their progress, estimate waiting times, and notify them through WhatsApp or web push when their turn is approaching. Staff will use role-based dashboards to call patients, transfer them between departments, update service status, and monitor congestion. Hospital administrators will use reports and analytics to identify bottlenecks, reduce waiting times, improve staff allocation, and make evidence-based decisions. The system will also support low-bandwidth and local network deployment, making it suitable for public hospitals in Zimbabwe and other resource-constrained environments.

CORE PATIENT JOURNEY

01	02	03	04	05	06
Register	**Triage**	**Consult**	**Diagnose**	**Treat**	**Complete**

DESIGNED TO WORK

LOCAL	CLOUD	HYBRID
Hospital LAN and local server	Remote access and central reporting	Local continuity with cloud backup

WAITLESS VISION **A safer, faster and more transparent patient journey - supported by technology that works in real hospital conditions.**